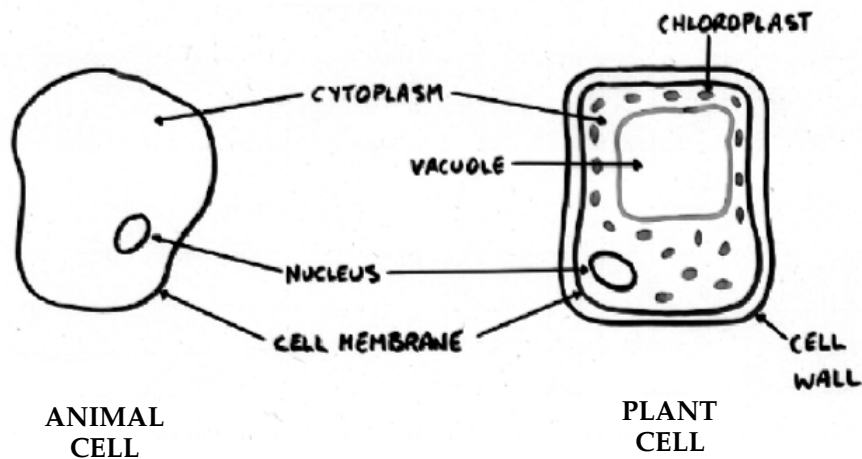


CELL STRUCTURE



Features Common To All Cells

- **Nucleus** - control centre of the cell; controls all activities such as growth and reproduction of the cell.
- **Cell membrane** - thin layer which holds all the cell contents together; allows O_2 , nutrients and H_2O to pass in and out of the cell.
- **Cytoplasm** - all living material inside a cell but outside the nucleus.

Differences Between Plant And Animal Cells

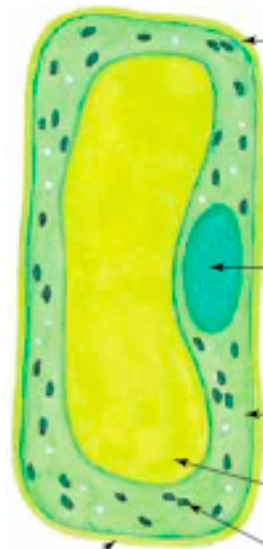
Plant cells have the following features. Animal cells don't.

- **Vacuole** - water storage compartment which helps maintain water balance in a plant.
- **Chloroplasts** - contain green chlorophyll which is involved in photosynthesis (the process of making food).
- **Cell wall** - rigid wall made of cellulose; gives a plant its shape.

Note

Animal cells don't have a rigid cell wall so that they can alter their shape as an animal moves.

Plant cell



Cell wall

The tough covering around plant cells is the cell wall. It gives plant cells strength and holds them in shape. Plant cell walls are made of a substance called **cellulose**. Water and dissolved substances can pass through the cell wall. Animal cells do not have a cell wall.

Cell membrane

The thin layer that encloses the cytosol is the cell membrane. It keeps the cell together and gives it its shape. Some substances, such as water and oxygen, can pass through the cell membrane, but other substances cannot. The cell membrane controls what enters and leaves the cell.

Nucleus

The nucleus is the control centre of the cell. It contains DNA in the form of chromosomes and controls what the cell does and when.

Cytosol

The jelly-like substance inside cells is the cytosol. It contains many important substances, such as glucose, that are needed for chemical reactions that occur inside cells.

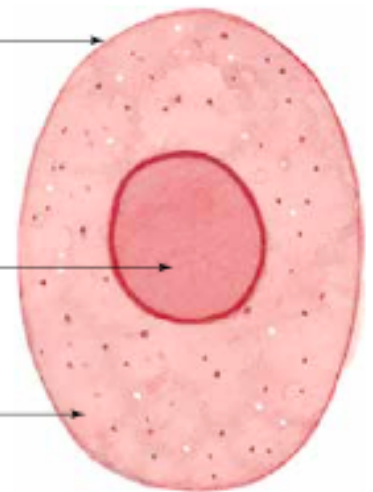
Chloroplasts

Chloroplasts are the oval-shaped organelles found only in plant cells. Chloroplasts contain a green substance called chlorophyll. Chloroplasts use energy from the sun to make food. Not all plant cells contain chloroplasts. They are found only in leaf and stem cells.

Vacuole

The vacuole is an organelle used to store water and dissolved substances. Vacuoles can look empty, like an air bubble. Plant cells usually have one large vacuole. The mixture inside a plant's vacuoles is called **cell sap**. The red, blue and violet colours that you often see in plant leaves and flowers are due to the substances stored in vacuoles. Most animal cells don't have vacuoles.

Animal cell



Some differences in the basic cell design in the five kingdoms

Characteristic	Kingdom				
	Animalia (animals: e.g. lizards, fish, spiders, earthworms, sponges)	Fungi (e.g. yeasts, moulds, mushrooms, toadstools)	Plantae (plants: e.g. ferns, mosses, conifers, flowering plants)	Protoctista (e.g. algae, protozoans)	Prokaryotae (bacteria and cyanobacteria)
Number of cells	Multicellular	Usually multicellular but some unicellular	Most multicellular	Unicellular or multicellular	Unicellular
Nucleus	Present	Present	Present	Present	Absent
Cell wall	Absent	Present	Present	Present in some	Present
Large vacuole	Absent	Absent	Present	Present in some	Absent
Chloroplasts	Absent	Absent	Present in leaf and stem cells	Present in some	Absent (but chlorophyll may be present in some)